

Ruturaj Dilip Gawade

Automation and Commissioning Engineer

PLC | HMI | SCADA | Commissioning | Wiring and IO Validation | AutoCAD Electrical | Customer Support

About Me

I am an Automation and Commissioning Engineer with 3 years of practical experience in industrial automation. My work has taken me through automotive manufacturing, conveyor and lift systems, boiler and utility automation, food processing, pharma production, multi-line packing plants, and SCADA-heavy assembly systems. What I enjoy most about this field is that it never stays limited to one screen or one software. A sequence that looks correct in a program still has to survive field noise, wiring issues, timing mismatches, operator pressure, and production expectations. That reality shaped the way I work.

My interest in automation began during my diploma studies when I first saw ladder logic controlling a real robot. That moment made the field feel alive to me. From there I moved into industrial projects where I learned step by step how machines really behave. I learned programming, but I also learned the parts that are easy to ignore in a simple summary: how sensors fail, how one wrong permissive can stop a machine, why clean wiring matters, why operators need practical alarms, and why calm troubleshooting matters during critical restarts. Over time I grew from supporting tasks to taking responsibility for development, commissioning, customer coordination, and project handover.

Professional Journey

At Promatics Solutions in Pune, from March 2021 to April 2024, I worked on projects from the initial requirement discussion to final handover and later support. My day-to-day work included PLC programming, HMI and SCADA development, communication setup, sequencing, alarm logic, IO checks, loop testing, analog signal validation, sensor and actuator troubleshooting, manual recovery logic, and site commissioning. I also supported FAT, SAT, documentation, backups, and operator training, so the machine was not only delivered but also understood by the people who would run it.

One thing that clearly stands out in my portfolio is that my work was not only software focused. In several projects I supported field wiring verification, terminal and IO mapping, wiring mismatch correction, and coordination with electrical teams. I also worked with AutoCAD electrical drawings, updating documents to match actual machine changes during assembly and commissioning. On top of that, I regularly handled customer communication before development, during trials, and after go-live. That taught me how to translate technical issues into simple decisions, stay calm under pressure, and make fixes that are safe and lasting instead of temporary patches.

What My Work Usually Includes

Work area	What that meant in real projects
Development	PLC logic, HMI screens, SCADA monitoring, interlocks, alarms, communication mapping, diagnostics, and clear Auto and Manual behavior that operators can actually use.
Commissioning	IO checks, loop checks, dry runs, hot trials, calibration support, analog scaling validation, timing correction, and real shop floor troubleshooting during production pressure.
Electrical support	Field wiring verification, terminal and IO mapping, mismatch correction, coordination with electricians, and AutoCAD Electrical drawing updates when the built machine changed.
Ownership	Customer discussions before development, daily updates during commissioning, technical handover, backups, operator training, and remote after-sales support through AnyDesk or VPN.

Selected Project Highlights

Automotive Test Loop Line Automation

Client origin: Germany | Role: PLC and HMI Programmer, Commissioning Support | Tools: Mitsubishi FX5U, GX Works 3, GT Designer 3

This automotive line was more than one machine. It was a full test loop where conveyors, turntables, presses, stoppers, and lift tables had to keep handing work to one another without losing rhythm or creating unsafe movement. My responsibility was to make the logic feel predictable. I built step-based sequences, added feedback checks before motion, and kept Auto and Manual modes clearly separated so operators could recover safely instead of fighting the machine. I also designed alarms and status screens that answered practical questions quickly. During commissioning I validated sensors and actuators, corrected timing issues, and helped the customer team move from trial conditions to stable running with more confidence.

Boiler House PLC and SCADA Automation

Client origin: India | Role: PLC and SCADA Programmer, Commissioning Engineer | Tools: Siemens S7-1200, TIA Portal V14, WinCC SCADA

The boiler house project demanded process stability, not just machine sequencing. Fuel handling, dual furnaces, draft fans, and the feed-water system all had to behave safely and in coordination. I developed PLC logic for start-up and shutdown, worked on analog input and output scaling, and supported PID-based control for steam pressure and tank level so the process remained smooth under real plant conditions. On the SCADA side, I created trends, alarms, and operator views that made the plant easier to understand during hot trials. I also supported loop checks, trial running, and customer handover, because utility systems are trusted only when operators feel that the plant reacts clearly and safely.

Multi Lift Conveyor System Integration

Client origin: United Kingdom | Role: PLC and HMI Programmer, On-Site Commissioning Engineer | Tools: Allen-Bradley CompactLogix, Studio 5000, FactoryTalk View ME

This project involved integrating new lift stations into an already running conveyor and turntable system. That made the work more sensitive, because the new logic had to fit naturally into an existing flow without disturbing the rest of the system. I modified sequence logic, built stable request and acknowledge handshakes between sections, and worked on encoder-based lift positioning with homing, limits, and position validation. I also created HMI views that made Auto and Manual behavior easier to follow on site. During commissioning I validated feedback signals and station interfaces under tight timelines, coordinated closely with the customer team, and supported training so the final solution felt reliable rather than newly patched.

Tea Production Line Process Automation

Client origin: India | Role: PLC and SCADA Engineer, Commissioning Support | Tools: Siemens S7-1200, TIA Portal V16, WinCC Runtime Advanced

In the tea production line, I worked across feeding, drying, fermentation, blending, and packing. The challenge was to keep the system modular enough to test section by section, while still ensuring that the overall process behaved as one connected flow. I developed PLC logic for conveyors, feeders, process equipment, and packaging sequences, and worked on temperature, humidity, and airflow control using PID loops with practical limits for production. I also created WinCC screens, alarms, trends, and reporting views that helped operators understand the process rather than only monitor it. Alongside programming, I supported calibration, commissioning, and customer coordination so the system stayed practical for everyday use.

Press Machines Automation with MES

Client origin: United States | Role: PLC, HMI and Electrical Design Engineer, Commissioning Support | Tools: Siemens S7-1200, KTP700 HMI, TIA Portal V16, AutoCAD Electrical

This project is important in my portfolio because it clearly shows that my work goes beyond software alone. I handled two press machine workflows where safe sequencing, part presence logic, servo and stepper positioning, and repeatable cycle behavior were all essential. Along with PLC and HMI development, I supported MES data integration so production status could be captured more clearly. I also worked with AutoCAD electrical drawings and helped keep documentation aligned with actual machine changes during development and commissioning. That combination of control logic, traceability thinking, and electrical documentation reflects how I usually work on projects where mechanical, electrical, and software changes all meet in the same machine.

Stud Pressing Line OPC UA Upgrade

Client origin: Italy | Role: PLC and HMI Programmer, On-Site Commissioning Engineer, After-Sales Support | Tools: Siemens S7-1200, TIA Portal V14, OPC UA

Here the plant needed two things at once: an upgrade for a new press model and better visibility through OPC UA-based monitoring. Because it was an existing line, success depended on understanding the real machine first before changing it. I modified PLC and HMI logic so the sequence remained predictable, safety curtain related behavior stayed clear, and Manual recovery did not become risky. I also verified wiring and IO behavior on the shop floor, mapped useful data such as counters, alarms, and cycle status for central monitoring, and later supported after-sales troubleshooting with structured diagnostics. This project showed my comfort with improvement work, where careful modification matters more than fresh development alone.

Spice Grinding and Packing SCADA System

Client origin: India | Role: Automation and Commissioning Engineer | Tools: Siemens S7-1500, TIA Portal V14, WinCC Runtime Advanced

This multi-line spices plant project became memorable because it turned into an urgent production recovery case. Five different grinding and packing lines had to be monitored through one SCADA station, but the immediate challenge was that the plant needed to restart quickly after a breakdown. I worked directly with field signals, VFD behavior, alarm linking, communication checks, and emergency wiring corrections where required. The goal was not to make the system look good on screen first. The goal was to restore safe and stable production. Once the basics were corrected, the centralized SCADA view became genuinely useful because operators could see all five lines clearly and react faster when something changed.

Multi Station Assembly SCADA Optimization

Client origin: Italy | Role: SCADA Development and Integration Engineer | Tools: ASEM IPC and HMI, CSV alarm workflow, OPC, TIA Portal V17

In this project, the line already existed, but the SCADA layer was hard to live with. Alarm handling was heavy, tag structures were inconsistent, and troubleshooting was slower than it should have been. I rebuilt a large alarm database of around 1800 alarms using CSV export and import, cleaned tag structures across four stations and the supervisory PC, improved synchronization, and added recipe-based control to support easier changeovers. I also validated the end-to-end behavior with the customer team so the final result was not only technically cleaner but also easier for operators and maintenance staff to use daily. This work showed my strength in improving readability and maintainability in complex supervisory systems.

Pharma Powder Handling Line Commissioning

Client origin: India | Role: Automation and Commissioning Engineer | Tools: CompactLogix 5370, Studio 5000, PanelView Plus 7, FactoryTalk View ME

This pharma project involved conveying, grinding, mixing, and packing, so the system had to remain stable, readable, and maintenance-friendly. I validated IO, supported calibration of analog transmitters, stabilized Auto cycles, integrated VFD control with setpoints and feedback, and developed HMI screens that clearly showed blocked conditions and safe recovery paths. In pharma environments, vague alarms and unstable behavior can quickly become serious operational issues, so I paid attention to diagnostics and structured sequence control. I also supported customer coordination, operator training, and final handover documentation. The result was a system that was not only commissioned successfully, but also easier for the plant team to run and troubleshoot after handover.

Working Style in the Field

I approach projects from the field backwards. Before trusting a program, I like to understand what the machine is really doing at the sensor level, what the operator actually sees, and where maintenance teams usually struggle. That habit helped me a lot during commissioning and breakdown support, because many issues that look like software problems actually begin in wiring, feedback, scaling, timing, or unclear recovery logic. I try to connect all of those layers instead of looking at only one.

I also value the human side of engineering. During critical trials or production stops, customers do not only need a technical answer. They need a calm person who can explain what happened, what will be changed, and why the fix is safe. That is why I pay attention to documentation, clear alarms, operator guidance, and practical handover. My goal has always been the same: build systems that run well, are easy to understand, and stay maintainable after the excitement of commissioning is over.

Education

- Master in Digital Engineering, Otto von Guericke University Magdeburg, Germany | 04.2024 to Present
- Bachelor of Technology in Electrical Engineering, Savitribai Phule Pune University, India | 06.2020 to 07.2023
- Diploma in Electrical Engineering, Maharashtra State Board of Technical Education, India | 07.2018 to 03.2020

Skills

PLC Hardware: Siemens S7-1200; Siemens S7-1500; Siemens Safety F-Series; Allen-Bradley CompactLogix; Allen-Bradley ControlLogix; Mitsubishi FX5U; Beckhoff; Schneider Electric; Omron; Delta; ASEM; Industrial PCs

Software and SCADA: TIA Portal V14 to V17; STEP7; WinCC SCADA and Runtime Advanced; Studio 5000; FactoryTalk View ME; GX Works3; GT Designer3; TwinCAT 3; AutoCAD Electrical

Industrial Networking and Data: OPC UA; Profinet; EtherNet/IP; Modbus TCP; Profibus; MES data integration

Field Automation and Control: Motion control for servo and stepper systems; VFD integration; PID loop tuning; safety interlocks and light curtains; alarm and recipe management; FAT and SAT execution; IQ and OQ validation protocols

Process Expertise: Automotive assembly and press lines; conveyor and lift automation; boiler and utility automation; process industry; food, pharma, and packaging systems

Final Note

This summary is short by design, but the work behind it has always been hands-on and close to real production needs. The strongest pattern across my portfolio is not only the range of software or industries. It is the ability to move between development, field execution, electrical reality, operator needs, and customer expectations without losing clarity. That is the kind of engineering work I want to keep building on.

Contact

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